

KEERTHI KUMAR KARANI

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PROFESSIONAL SUMMARY:

- IT Professional with **8 years of experience** in Data Engineering and Analytics architecting Azure Databricks Lakehouse platforms using Delta Lake Medallion Architecture across insurance, healthcare, telecom, and HRMS domains.
- Architected end-to-end **data ingestion pipelines** processing 1M+ daily transactions with sub-minute availability through real-time and batch ingestion using PySpark and Spark SQL from Azure Event Hubs, Kinesis, and partner APIs.
- Designed **medallion-pattern** data lake and Lakehouse platforms on Azure and AWS — consolidating fragmented source systems into unified bronze/silver/gold Delta layers with fact and dimension schemas, eliminating ~60% of enterprise data silos.
- Optimized **Delta Lake query performance** using Z-ORDER, OPTIMIZE, partition pruning, and predicate pushdown improving Databricks SQL dashboard response times by 35–40% and reducing compute overhead.
- Implemented **CDC and incremental loading** patterns on Databricks, eliminating full table reloads and accelerating pipeline processing by up to 45% across SLA-critical workflows.
- Enforced HIPAA, PCI-DSS, GDPR, and SOC 2 **compliance** using Unity Catalog, Databricks RBAC, row-level security, dynamic data masking, and Azure Key Vault- zero data breaches across 3 audit cycles.
- Established data lineage tracking, metadata management, and business glossaries using **Unity Catalog and Azure Purview** - improving data discoverability by 60% across regulated pipelines.
- Executed **enterprise-scale data migrations** (Teradata-to-Snowflake) and multi-source system consolidations using PySpark, AWS Glue, and Databricks across telecom and HRMS domains.
- Streamlined **deployment cycles** by ~35% through end-to-end CI/CD pipelines via Azure DevOps and Git with automated notebook testing, version control, and environment promotion.
- Led onshore/offshore teams** of 5–8 engineers across 5 industry domains, translating complex business requirements into scalable data ingestion, pipeline, and governance solutions.

TECHNICAL SKILLS:

Core Stack	Azure Databricks, Delta Lake, PySpark, Spark SQL, Medallion Architecture, Databricks SQL, Databricks Jobs, Unity Catalog, Azure Data Factory
Languages	Python (PySpark, Pandas, NumPy), SQL, Unix Shell Scripting
Data Engineering	Databricks, Delta Lake, PySpark, ETL Pipelines, Apache Kafka, Airflow, Spark, Data Quality Frameworks, Data Migration, Batch & Near Real-Time Data Processing, Hive
Cloud - Azure	Azure Data Factory, Azure Event Hubs, Azure Synapse Analytics, ADLS Gen2, Azure Key Vault, Azure Logic Apps, Azure Functions, Azure DevOps, Azure Monitor, Cosmos DB

Cloud - AWS	S3, EC2, Redshift, AWS Glue, AWS EMR, Kinesis, Lambda, Step Functions, CloudWatch, Amazon QuickSight
Data Visualization & Analytics	Power BI, Tableau, Excel (Advanced), Power Query, KPI Reporting, Trend Analysis, Dashboard Development
Statistical & Analytical Methods	Regression Analysis, Time Series Analysis, Hypothesis Testing, A/B Testing, Exploratory Data Analysis
ETL Tools	SSIS, Informatica, Talend, DataStage, AbInitio
Databases	Snowflake, Teradata, Oracle, Microsoft SQL Server, MySQL, PostgreSQL, Db2, T-SQL, Query Optimization, Stored Procedures, Functions
Cloud Services	Azure: Data Factory, Synapse Analytics, Cosmos DB; AWS: S3, EC2, Redshift, AWS Glue, Amazon QuickSight
Tools & Software	Git, Jenkins (CI/CD), Jira, Bitbucket, Erwin, Lucid Chart, Azure DevOps, , Docker
Security & Compliance	RBAC, Data Encryption, Azure Key Vault, HIPAA Compliance
Functional Skills	Predictive Analytics, Data Analysis, Data Visualization, Time Series Forecasting, Data Modeling, Data Pipeline Design, Data Integration, Data Management
Operating Systems	Windows, Linux/Unix-like systems

CERTIFICATIONS:

Azure Certified, Fundamentals

Azure Certified, Data Fundamentals

Azure Certified, Fabric Analytics Associate

Azure Certified, Fabric Data Engineer Associate

Google, Engineer Data in Google Cloud

PROFESSIONAL EXPERIENCE:

Client: Assurant FL

Jul'25 – Present

Role: Senior Data Engineer

Responsibilities:

- Eliminated ~60% of cross-brand data silos by architecting a **Databricks Lakehouse** using Delta Lake Medallion Architecture (bronze/silver/gold) with fact and dimension schemas in the gold layer, consolidating claims data across 3+ business units for unified fraud detection.
- Engineered real-time and batch **PySpark and Spark SQL ingestion pipelines** consuming warranty claims and policy data from Azure Event Hubs and 10+ partner APIs with sub-minute data availability.
- Accelerated claims adjudication from 72 hours to under 36 hours through **CDC and incremental loading** patterns on Databricks, eliminating full table reloads and cutting pipeline time by ~45%.

- Strengthened claim approval accuracy by deploying a **data quality validation framework** using Databricks Expectations and Python rules, cutting downstream reporting errors by ~30%.
- Compressed quarterly compliance reporting from 5 days to under 8 hours by curating gold-layer **Delta tables** and automating transformation workflows from raw claims and policy data.
- Optimized high-volume claims tables (50M+ records) using **Z-ORDER, OPTIMIZE, and partition pruning** — improving Databricks SQL dashboard response by ~40% and reducing cluster compute overhead.
- Delivered **Databricks SQL dashboards** tracking claims volumes, device protection KPIs, denial rates, and fraud indicators — enabling leadership to spot fraud patterns 3x faster.
- **Safeguarded multi-brand insurance programs** with zero unauthorized access incidents across 3 audit cycles using Databricks RBAC, Unity Catalog, and Azure Key Vault under PCI-DSS standards.
- Implemented **data lineage tracking** and impact analysis using Unity Catalog, ensuring full auditability of financial data transformations for regulatory compliance across insurance verticals.
- **Enforced PII/PHI protection** through dynamic data masking and data classification policies in Unity Catalog and Azure Purview, meeting PCI-DSS and state insurance regulatory requirements.
- Established semantic data layers and business glossaries in **Unity Catalog**, creating standardized data contracts across claims, warranty, and policy domains for self-service analytics.
- Sustained 99.5%+ pipeline reliability through **Databricks Jobs** with dependency chaining, automated retries, and Azure Monitor alerting across ingestion and transformation workflows.
- Streamlined deployment cycle time by ~35% via **Azure DevOps CI/CD** with automated Databricks notebook testing, version control, and environment promotion across dev/UAT/prod.
- Served as **onsite lead managing an offshore development team of 4+ engineers** — conducting code reviews, design reviews, and solution walkthroughs to align offshore delivery with architecture standards and business requirements.
- Authored production-quality, maintainable **PySpark and Spark SQL transformation code** — orchestrated via Azure Data Factory and Databricks Jobs — to build and maintain gold-layer lakehouse models for claims risk scoring and warranty analytics.

Environment: Azure Databricks, Delta Lake, PySpark, Spark SQL, Unity Catalog, Azure Purview, Azure Event Hubs, Azure Data Factory, Azure Functions, ADLS Gen2, Databricks SQL, Databricks Jobs, Azure Key Vault, Azure Monitor, Azure DevOps, Git, CDC, Z-ORDER, OPTIMIZE, Python, CI/CD, PCI-DSS

Client: Community Dreams Foundation

Sep '23 – Jul '25

Role: Data Engineer

Responsibilities:

- Unified 8+ fragmented HR source systems into a Databricks Lakehouse with **Delta Lake, PySpark, and Spark SQL** ingestion pipelines and fact/dimension gold-layer schemas — resolving chronic payroll-to-headcount mismatches that had caused quarterly correction cycles.
- Elevated payroll processing accuracy to ~98.5% by designing **dimensional models** across bronze, silver, and gold Delta layers with full audit traceability — eliminating manual reconciliation errors.
- Slashed BI team ticket volume by ~40% through optimized **Databricks SQL queries** and reusable parameterized views, cutting query response time by ~35% and shifting the org toward self-service.
- Tuned **Delta Lake performance** on high-volume HR tables using Z-ORDER, OPTIMIZE, and partition pruning — improving Databricks SQL dashboard refresh times and reducing job cluster costs.
- Surfaced early-warning attrition signals HR previously missed by publishing curated gold-layer Delta datasets to **Power BI semantic models** — improving prediction accuracy by ~25%.
- Prevented recurring payroll discrepancies by building **automated PySpark validation pipelines** using Databricks Expectations that flagged anomalies before downstream consumption.

- Centralized data cataloguing, metadata management, and lineage tracking using **Unity Catalog** — satisfying SOC 2 auditors without manual evidence gathering and **improving discoverability by ~60%**.
- **Orchestrated complex data workflows** using Apache Airflow and Azure Data Factory — managing dependencies across payroll ingestion, benefits reconciliation, and headcount reporting, while coordinating offshore engineers through design reviews and solution walkthroughs to ensure platform quality.
- **Monitored** Databricks Jobs alongside Airflow DAGs for transformation cycles, giving operations teams centralized pipeline health visibility and cutting incident resolution time by ~40%.
- Secured **sensitive HR data** using dynamic data masking and column-level security in Unity Catalog — meeting SOC 2 and organizational PII standards.
- Instituted **data classification frameworks** categorizing HR datasets by sensitivity level (public, internal, confidential, restricted) — automating policy enforcement and cutting compliance review time by ~45%.
- Authored standardized HR business glossaries in **Unity Catalog** (FTE, attrition rate, cost-per-hire), bridging the language gap between engineering and HR teams.
- Pinpointed recruitment funnel bottlenecks by developing **workforce dashboards** tracking headcount trends and time-to-hire KPIs — enabling HR leadership to cut time-to-hire by **~20%**.
- Partnered with HR operations and finance to design curated **workforce planning datasets**, improving headcount forecasting accuracy by ~30% for proactive budget allocation.

Environment: Azure Databricks, Delta Lake, PySpark, Spark SQL, Unity Catalog, Apache Airflow, Azure Data Factory, Power BI, ADLS Gen2, Databricks Jobs, Databricks SQL, Z-ORDER, OPTIMIZE, Azure Key Vault, Azure Purview, Azure Monitor, Python, Azure DevOps, Git, CI/CD

Client: AT&T, DirecTV- (AMDOCS)

Nov '21 – Jun '23

Role: Business Intelligence Developer

Responsibilities:

- Spearheaded a **Snowflake marketing data hub** on AWS supporting 1M+ daily transactions across AT&T and DirecTV brands, improving data accessibility by 30% and aligning SLAs with business stakeholder requirements.
- Developed **PySpark ETL pipelines** on AWS EMR transforming telecom customer data for Snowflake ingestion, supporting real-time marketing segmentation and campaign performance analytics at scale.
- Executed a **Teradata-to-Snowflake migration** using AWS Glue and PySpark, reconciling 3+ years of historical data and cutting migration runtime by 40% via optimized parallel processing and automated reconciliation scripts.
- Designed **star-schema dimensional models** in Snowflake using AWS Glue-processed datasets, improving Power BI dashboard query performance by 35% and reducing report load times from minutes to seconds.
- Configured **CDC and incremental load strategies** using AWS Database Migration Service (DMS) and AWS Glue, reducing data latency by ~50% versus the prior batch-only approach for customer account synchronization.
- Built scalable data ingestion workflows using AWS Kinesis Streams to process real-time telecom events and customer transactions, enabling sub-second data availability for **marketing and operational analytics**.
- Resolved recurring EMR pipeline failures through root cause analysis — identifying data skew and inefficient joins, reducing failures by 20% and saving ~6 hours/week of incident management overhead.

- Engineered **ML-ready churn feature datasets** using AWS Glue and PySpark transformations, stored and versioned on AWS S3 — enabling prediction models that reduced customer churn by 15% and accelerated insight delivery by 40%.
- Implemented **data lake architecture on AWS S3** organizing raw, curated, and analytics-ready telecom datasets using medallion pattern, with AWS Glue Data Catalog for metadata management and data discoverability.

Environment: AWS EMR, AWS Glue, AWS S3, AWS Kinesis, AWS DMS, AWS Redshift, PySpark, Snowflake, Power BI, Teradata, GitHub, Jenkins, Erwin, Lucid Chart

Client: United Health Group, T-Mobile – (Infosys)

May '18 – Oct '21

Role: ETL Developer

Responsibilities:

- Led dimensional modeling and optimization of a **Teradata enterprise data warehouse** for United Health Group, improving retrieval efficiency by 30% through partitioning, indexing, and HIPAA-compliant governance controls.
- Designed **HIPAA-compliant EHR/EMR dashboards** using Power BI, Tableau, and Excel, increasing reporting efficiency by 35% and eliminating ~8 hours/week of manual report production.
- Built **real-time healthcare monitoring dashboards** using Azure Synapse Analytics and Power BI, reducing clinical leadership decision-making latency by 40% through proactive KPI monitoring.
- Developed **T-Mobile customer churn dashboards** and predictive feature datasets using Power BI and Python, enabling retention campaigns that reduced customer churn by 15%.
- Implemented **HIPAA-aligned governance with Azure RBAC** and field-level encryption in ADF and Azure Data Lake pipelines, reducing operational compliance errors by 20% through root cause analysis and regression testing.
- Developed **HL7, ORU, and ADT message ingestion workflows**, ensuring seamless integration with hospital information systems and downstream clinical analytics platforms.
- Automated **ETL job monitoring and data validation** using Python scripts, reducing manual oversight by 25% and improving pipeline reliability across multi-source healthcare data environments.

Environment: Python, SQL, Power BI, Tableau, Teradata, Snowflake, Azure Synapse, Azure Data Factory, ADLS, SSIS, SQL Server, Unix, Jenkins, HIPAA, RBAC